Personal Finance Advisor v3.4: how the flow works, and how to demo it

Demo in one line: use the app like a real user, then open n8n's **Executions** tab to show what each request ran. Test mode (`Execute workflow` , `scripts/test-step.sh`) is a debugging tool for one step at a time. You don't need it for a demo.

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1. The big picture: one app, seven doors

The advisor is a **web app served by n8n**. Everything the user does in the browser calls one **webhook**, a starting node in the workflow. Each call is a separate run, listed in **Executions**.

#	The user…	Webhook (starting node)	What happens
1	opens `localhost:5678/webhook/pfa`	**Advisor App** (GET)	n8n sends the app page
2	types their email	**Lookup API** (POST)	finds the saved profile and past plans
3	fills the 5 steps and presses **Submit**	**Submit API** (POST) ★	**the main pipeline**, section 2
4	sees "Building your plan…"	**Status API** (GET)	the page asks "is it ready?" every few seconds
5	sees the report open	**Report API** (GET)	loads the saved report and draws the charts
6	asks a question in the chat	**Chat API** (POST) ★	**the chat agent**, section 3
7	clicks **Yes / No** on a plan change	**Confirm API** (POST)	Yes runs the main pipeline again with the change

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2. The main pipeline (Submit API)

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Submit API
  ├ Plan Request (form)            tidy the submitted answers
  ├ Build Financial Profile        "18L" → 1,800,000; pay, EMIs, surplus
  ├ Goal Text Guardrail (LLM)      blocks prompt injection in free text
  ├ Check Profile Completeness
  ├ Needs More Info? ── missing ──► Explain Fixes ► Respond: Fix These
  │        │ ok
  ├ Respond: Building              browser answered now; rest runs in background
  ├ Prepare User Storage ► Upsert User Profile (Postgres)
  ├ Restore User State
  │    ├──► Compute Debt Metrics       ► Debt Analysis Agent (LLM)      ┐ in
  │    └──► Compute Allocation Metrics ► Portfolio Analysis Agent (LLM) ┘ parallel
  ├ Sync Analyses (merge)
  ├ Compute Future Projection      goal reached at this saving rate?
  ├ Low risk → bank deposits? ── yes ──► Plan Bank Deposits
  │                                      ► Attach Deposit Plan
  ├ Investment Recommender Agent (LLM + tools)
  ├ Assemble Report Inputs
  ├ Report Generator Agent (LLM, structured JSON) ► Render Report Card
  └ Save Advice History (Postgres) report saved; Status API says "ready"
```

Investment Recommender Agent tools:

- Knowledge_Base_Search (RAG on pgvector)
- Historical_Market_Data
- Screen_NIFTY50_Stocks
- Company_Annual_Performance
- Plan_Bank_Deposits
- Get_Deposit_Rates

Key design point: the **numbers come from code nodes** (deterministic). The **LLM agents only explain and recommend**, using those numbers, and are told never to recalculate them. That's what keeps the report's maths reliable.

(The *Append to Google Sheets* node is switched off on purpose.)

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3. The chat (Chat API, or n8n's "Open chat")

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Chat API ► Chat Report Owner ► Chat Context ► Chat Input Guardrail
   ► Financial Q&A Agent
   ► Chat Output Sanitizer ► Chat Result
   ► Plan Change? ── yes ──► Yes / No buttons ► Confirm API
```

Financial Q&A Agent tools:

Tool group	Tools
This user's data	the profile and latest report
Search	RAG search
Market data	market history, NIFTY50 screener, company results
Deposits	deposit planner and rates, India inflation
Teammate's service	Ask_Bank_Deposit_Schemes, Ask_Mutual_Funds
Plan changes	Propose_Plan_Change

Where the teammate's project fits: questions about a bank's scheme rules or a specific mutual fund go to the teammate's container, which runs their Chroma RAG over bank and fund documents and is called as a tool.

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4. Five-minute demo script

Before the meeting

- Run `./start.sh`, then `./healthcheck.sh`. Every line should say OK.
- Open two browser tabs:
  - the **app**: `localhost:5678/webhook/pfa`
  - **Executions**: `localhost:5678/workflow/pfaAdvisorV32App/executions`

1. The app (2 min). Enter a new email and fill the 5 steps with TESTING.md scenario A1:

- CTC 18L, goal *Buy a home*, 80L in about 5 years
- expenses 45K, car loan 3.2L at 9.2% with an 11K EMI
- MF 4.5L with a 15K SIP, FD 2L
- risk Medium

Press **Submit**. While "Building your plan…" shows, say: *"n8n has already answered the browser. The agents are running in the background."*

2. n8n while it builds (1 min). In Executions, open the **Submit API** run.

- Show the green path: guardrail, profile, **two agents in parallel**, the recommender with its tools, the report.
- Click **Investment Recommender Agent** → **Logs** to show the RAG search and the tool calls.

3. The report (1 min). Charts, the risk-profile card, "where to put your money each month", the safety-net meters.

4. The chat (1 min).

- Ask *"Which bank gives the best 1-year FD for me?"* to show tools at work.
- Then ask *"My expenses went up to 50K."* The **Yes / No plan-change** buttons appear. Open that run in Executions as well.

Optional: Phoenix at `localhost:6006`, project **pfa-n8n**, shows every LLM call with its prompt, tokens and cost.

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5. Likely mentor questions

Question	Answer
Why webhooks, not n8n's Form node?	A custom multi-step app: Back / Edit, pre-filled answers for returning users, live checks as you type.
How do you stop made-up numbers?	Code computes every number, agents only explain them, and outputs are sanitised.
How is it kept safe?	Guardrails on the goal text and on chat input, and a sanitiser on chat output.
Where is the data?	Postgres: profiles and report history, plus RAG embeddings in pgvector.
How do you see what the AI did?	n8n Executions → Logs on each agent, and Phoenix traces with tokens and cost.
Why does the report take 1-3 minutes?	Four LLM agents and tool calls. The page polls Status API, so the browser never times out.